Input

Full Multimodal Evidence X

Aa

1. Video-Audio modality

Video Caption: dogs playing



Audio Caption: bird chirping

Question

Is there a bird visible in the video?

2. Image-Text modality



After a long game of fetch with its red ball, the dog curled up beside its stuffed toy to rest.

Question

What is the dog resting beside?

3. Text modality

(Objective) Postoperative delirium is common in older surgical patients. We investigated whether perioperative melatonin could reduce its incidence [...]
(Methods) 186 patients undergoing hip surgery were randomly assigned to receive melatonin or placebo [...]
(Results) The overall incidence of postoperative delirium was 19%. Delirium occurred significantly less often in the Melatonin group than in the Placebo group (12% versus 27%, p=0.01). Rates of serious adverse events were similar between groups [...]

Question

Does melatonin prevent postoperative delirium?

OmniLLM

Commitment Anchoring

Autoregressive Decoding

$T_1 \rightarrow T_2 \rightarrow T_3 \cdots T_n$

Joint Recoding

Candidate commitment y^{cand}

T_1 T_2 T_3 $\cdots$ T_n

Full-view Token Scores r^{full}

Evidence Interrogation

X-Text (Aa) | X-Image | X-Audio | X-Video

X^{-m}

Anchored Scoring

X^{-m} y^{cand} $\odot$ → r^{-m}

Evidential Dependence

$r^{full} - r^{-m}$ → $\delta^{(m)}$

Question × Token × Channel Confession

	Text	Image	Audio	Video
T_1	0	+	−	0
T_2	0	0	0	0
T_3	++	−	0	0

$\gamma(J^{cand})$

$\gamma(Free)$

0 : Aligned
− : Under-reliance
+ : Over-reliance
++ : Dominance

Initial Responses:

1. *Yes.*
2. *The dog is resting beside a ball.*
3. *No.*

Confession-Guided Correction

Judgement

$r(J^{cand})$ → $r(J^{cand})'$ → Reselect

$r(J^{alt})$ → Yes/No

Free form

Suspect

Revision Spans — Replaced Spans

Final Responses:

1. *No.*
2. *The dog is resting beside a toy.*
3. *Yes.*